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Facile fabrication of TiO_2-graphene nanocomposites (TGNCs) for the efficient photocatalytic oxidation of perfluorooctanoic acid (PFOA)

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ABSTRACT

The present work describes the facile synthesis of the TiO_2-graphene nanocomposites (TGNCs) graphene oxide-TiO_2 and reduced graphene oxide-TiO_2 and evaluates their photocatalytic performance and mechanism in the decomposition of intractable perfluorooctanoic acid (PFOA). More than 90% photocatalytic PFOA decomposition was achieved by the TGNCs (200 mg/L), while only 16% of the initial 12 μM PFOA was decomposed by the commercial TiO_2. The presence of graphene provided surface enhancement effects and limited the recombination of electron-hole pairs during photocatalytic decomposition. The effect of quenching agents, isopropyl alcohol and sodium persulfate, in the photocatalytic system gave insight into understanding the mechanism of decomposition.

1. Introduction

Perfluorinated substances (PFSs) are anthropogenic compounds that have been utilized in a wide range of applications since their first usage, which dates back nearly 70 years [1]. Due to their widespread usage, recalcitrance, persistence and toxicity, these compounds are ubiquitous in the environment and are of increasing concern [2–5]. Among the PFSs, perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) are of particular interest. These persistent organic pollutants are under surveillance by the European Union [6], the Stockholm Convention [7] and the US EPA [8]. However, PFSs are still found in the environment in considerable quantities [9,10]. The intractable nature of PFSs originates from the strong carbon-fluorine bonds present in the molecules [11]. The recalcitrance of PFSs has demanded that significant research be focused on their treatment, and as a result, noteworthy review papers have appeared in the recent literature [12–15]. The large quantity of treatment methodologies developed so far have yielded improved solutions for treating PFOA. However, the development of engineered materials for the effective treatment of PFOA is considered to be a suitable option and is still underway.

Heterogeneous photocatalysis using TiO_2 is consistently an interesting treatment option for the decomposition of emerging environmental contaminants [16,17]. The literature, however, shows that only limited performance is achieved when TiO_2 is used alone in the photocatalytic decomposition of PFOA [18–21], because of the large bandgap and rapid recombination of electron-hole pairs [20,21]. Modification of either the treatment process through the addition of chemicals like perchloric or oxalic acid [20,22] or the TiO_2 catalyst by incorporating transition metals like Pb, Fe or Cu [23,24] or noble metals like Pt, Pd or Ag [25,26] has been shown to improve the PFOA decomposition efficiency.

Another interesting advance recently made in the field of photocatalysis is the use of nanocomposites. Several studies have reported the photodecomposition of PFOA using various homogeneous and heterogeneous catalyst materials, including nanocomposites such as graphene-supported silicon carbide (SiC/graphene) [26], graphene-supported indium oxide (In_2O_3-graphene) [27], reduced graphene oxide (RGO)-supported zincoxide (ZnO/RGO) [28], noble metal-modified TiO_2 [25] and transition metal-modified TiO_2 [23].

Among the hybrid photocatalysts, the combination of TiO_2 with carbonaceous materials, particularly two-dimensional graphene, is an attractive approach due to the important attributes of these materials, such as a large specific surface area and high electron mobility. Furthermore, electron transfer from excited TiO_2 to graphene is defined by the interfacial electron transfer properties and strongly influences the photocatalytic activity of the nanocomposites [28]. Very recently,

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greater decomposition levels of up to 93 ± 7% were reported on a RGO/TiO_2 nanocomposite in the decomposition of 0.24 mmol/L PFOA in 12 h with 0.1 g/L catalyst under irradiation by a medium-pressure mercury lamp emitting light with a 200–600 nm wavelength [29]. The four-fold improvement in the decomposition efficiency of this photocatalyst relative to that of the TiO_2 photocatalyst was credited to the enhanced charge separation ability of RGO. Several reports have clearly documented the efficacy of various hybrid nanocomposites for the decomposition of PFOA, whereas the mechanism of decomposition remains unknown.

Based on the experimental results and the type of catalyst used, different decomposition mechanisms have been proposed in the literature for the decomposition of PFOA. A majority of the studies described that the PFOA decomposition was initiated by electron transfer from an adsorbed PFOA anion to a hole in the valence band of TiO_2, resulting in the formation of a perfluorocarboxy radical ($C_7F_{15}COO•$), which further underwent Kolbe decarboxylation to form compounds with shorter chain lengths [26–29]. The enhanced decomposition due to the noble metal-modified TiO_2 catalyst was attributed to the ability of the noble metal nanoparticles to act as electron sinks [26] or the individual roles played by In_2O_3 and graphene as an activity promoter and an electron sink, respectively in the case of In_2O_3-graphene catalyst [27], or perhaps due to the surface enhancement effects and the prevention of rapid electron-hole pair recombination during photocatalysis provided by the ZnO/RGO composites [28]. Sansotera et al. [30] studied the influence of oxygen gas on the photocatalytic decomposition of PFOA with commercial TiO_2 as a photocatalyst and suggested that the decomposition undergoes two-way oxidative paths involving photoredox processes and β-scission reactions.

The main objective of the present work was to utilize the advantages of GO and RGO in the application of these materials as supports for TiO_2 nanoparticles synthesized in the laboratory as well as for commercial TiO_2 (Degussa P25) in the photodecomposition of intractable PFOA under UV-C irradiation and mild reaction conditions. All the TiO_2-graphene nano composites (TGNC) photocatalysts were thoroughly characterized. Furthermore, the photocatalytic performance of the synthesized TGNCs was compared with that of the commercial catalyst, and the intermediates formed during PFOA decomposition were identified. The effects of quenching agents were also studied, which provided insight into the PFOA decomposition mechanism.

2. Materials and experimental methods

2.1. Chemicals

Graphite powder, L-ascorbic acid, PFOA, perfluoroheptanoic acid (PFHpA), perfluorohexanoic acid (PFHxA), perfluoropetanoic acid (PFPeA), perfluorobutanoic acid (PFBA), and perfluoropropanoic acid (PFPA) were purchased from Sigma Aldrich and used as received. HPLC-grade trifluoroacetic acid (TFA) and LC–MS-grade methanol were purchased from J.T.Baker. All the chemicals used in the present study were of analytical grade and used without further purification. Powder TiO_2 Degussa P25 was purchased from Aldrich with a purity > 95%.

2.2. Synthesis of GO

The synthesis of GO from natural graphite powder was achieved by a modified Hummers and Offeman method [31,32]. Briefly, 3 g of graphite powder, 6 g of $NaNO_3$, and 12 g of $KMnO_4$ were thoroughly dispersed in 120 mL of H_2SO_4 in a 250 mL round-bottomed flask. The reaction mixture was initially stirred in an icebath with a temperature maintained at 0–5 °C for 5 h. Thereafter, the temperature was slowly increased to 60 °C and maintained for 4 h. Later, the reaction mixture was cooled to room temperature. Next, 800 mL of water was added to this mixture, which was again cooled to 0–10 °C in an icebath. Then, 20 mL of 30% (w/v) H_2O_2 was added slowly, and the mixture was stirred for 1 h. The brownish solid formed was subsequently decanted and neutralized with repeated deionized water washings. The reaction mixture was centrifuged at 10,000 rpm and dried at 60 °C overnight in a hotair oven. The resulting GO was stored for further use.

2.3. Synthesis of RGO

Briefly, 250 mg of GO powder and 100 mL of water were mixed under sonication for 1 h. The mixture was then transferred to a 500 mL round-bottomed flask. To this mixture was added 400 mg L-ascorbic acid, and the solution was refluxed at 140 °C for 3 h. After completion of the reaction, the RGO product was allowed to cool to room temperature, and the settled RGO product was then collected by decantation. The final RGO product was washed several times with water and ethanol before air drying.

2.4. Synthesis of the TGNCs

The fabrication of 5 wt% GO-TiO_2 NCs was achieved by a sol-gel method. Typically, 245 mL of water and 5 mL of ethanol were mixed under stirring, and the pH of this mixture was adjusted to 3–4 with nitric acid to form the hydrolysis solution. To this 15 mL of ethanol and 5 mL of titanium tetraisopropoxide was added under stirring to prepare the precursor solution. The hydrolysis solution was heated to 60–80 °C under vigorous stirring, and the precursor solution was then added dropwise to form a gel. The solution was heated until the sol-gel reaction was complete, at which point a colloidal TiO_2 suspension formed. Later, 0.117 g of GO was added, and the mixture was sonicated for 30 min to produce a dispersion of GO sheets coated with TiO_2 nanoparticles. The resulting GO-TiO_2 colloidal suspension was allowed to dry at room temperature and then ground. Thereafter, the powder was subjected to calcination with a heating rate of 25 °C/min up to 450 °C and maintained at that temperature for 2 h. The final product is referred to as GO-TiO_2. In a similar experiment, GO was replaced with RGO, and the resulting powder is referred to as RGO-TiO_2. The fabrication of GO-P25 nanocomposites was achieved by a hydrothermal method. First, 2 mg of GO was mixed with 20 mL of DI water and sonicated for 30 min. To this aqueous dispersion of GO was added 10 mL of ethanol, and the resulting mixture was sonicated for 1 h. Later, 200 mg of commercial TiO_2 (P25) was added, and the resulting mixture was stirred for 2 h. Then, the TiO_2-containing mixture was transferred to an autoclave and subjected to heat treatment at 120 °C for 3 h to obtained the GO-P25 composite. After heating, the mixture was allowed to cool to room temperature and washed several times with water. Finally, the obtained GO-P25 composite powder was dried at 70–80 °C. The obtained catalyst materials were characterized using various techniques.

2.5. Characterization of the TGNCs

The phase structures of GO, RGO, GO-P25, GO-TiO_2 and RGO-TiO_2were studied by X-ray powder diffraction (XRD). The XRD measurements were performed using a Rigaku Miniflex-600 X-ray diffractometer operated with Cu Kα radiation (λ = 1.54 Å). The morphologies and microstructures of the nanocomposites were investigated by a TESCAN VEGA-3 scanning electron microscope (SEM) operated at an accelerating voltage of 30 kV. The compositions of the NCs were analyzed by energy dispersive X-ray spectroscopy (EDX) using an Oxford Instruments EDX detector attached to the SEM. The SEM and EDX of P25 and GO-P25 were obtained by using a JEOL JSM-7800 F thermal field emission scanning electron microscope. Furthermore, transmission electron microscopy (TEM) and high-resolution TEM (HRTEM) analyses of the TGNCs were performed using a JEOL JEM-2100 microscope operated at an accelerating voltage of 200 kV. The samples for both SEM and TEM analysis were prepared by ultrasonically dispersing a small amount of powder in ethanol and placing a drop of the diluted sample on a carbon-coated copper grid.

The sample-containing grid was air dried before performing the microscopy measurements. XPS analysis was performed with a VG Scientific ESCALAB 250 spectrometer. The specific surface area was calculated by the Brunauer-Emmett-Teller (BET) method from the nitrogen adsorption-desorption isotherm data using INSTRUMENT. The nitrogen adsorption-desorption isotherm plot for the RGO-TiO_2 at −196 °C can be seen in the supplementary data Figure S1. UV–vis spectrophotometer (JASCO V755) was used to record the absorption spectra of the photocatalysts.

2.6. Photocatalytic experiments

Photocatalytic PFOA decomposition experiments were carried out in a cylindrical quartz reactor with a reaction volume of 50 mL containing 200 mg/L photocatalyst (GO-TiO_2, RGO-TiO_2, GO-P25, or P25) and 12 μM PFOA while purging with 5 cc/min O_2 under irradiation by a UV-C lamp (Phillips), which had a wavelength of 254 nm with a power of 8 W and was housed in a quartz tube. The whole reaction solution was continuously stirred at 600 rpm on a magnetic stirrer. The temperature was maintained at 25 ± 2 °C. The pH value of the reaction during the experimental runs was not controlled in any of the experiments. The initial pH values in the test runs ranged from 4.64 to 4.95. The experiments were carried out for 8 h to evaluate the activity of the as-prepared NCs. Before starting irradiation, the mixture was stirred for 30 min under dark conditions to achieve the adsorption-desorption equilibrium. The photocatalytic decomposition efficiency of PFOA was studied under different GO-TiO_2 loadings, and the optimum load was employed in further experiments with the other catalyst materials in this study. Preliminary experiments on the adsorption efficiency of GO and the TGNCs such as GO-TiO_2 and RGO-TiO_2 for PFOA has been observed and the corresponding data has been presented in the Supplementary figure Figure S2. When we studied for the adsorption of 5 ppm PFOA on to 100 mg/L of GO in the solution, 45% of initial PFOA has disappeared from the solution. Whereas when a 100 mg/L of GO-TiO_2 or RGO-TiO_2 was used only a very little amount of initial 10 mg/L PFOA was observed to be removed from the solution. This could be because, in the case of TGNCs the percentage of TiO_2 was 95% as compared to 100% pure GO. It is well known that GO has high theoretical surface area and our previous studies has shown negligible adsorption of PFOA on to TiO_2. The light intensity measured at the outer surface of the quartz vessel with a UVP Multi-Sense MS-1 optical radiometer was 0.34 mW/cm^2. The samples were collected at specified intervals, filtered through a 0.45 μm PTFE filter and then analyzed by HPLC-MS/MS.

2.7. Analytical procedures

After filtering the sample aliquots through 0.22 μm Chrom Tech nylon syringe filters, the concentrations of PFOA and other shorter-chain PFCAs were measured by liquid chromatography–tandem mass spectrometry (LC–MS/MS). The analytical procedure for LC–MS/MS adopted in this study was described elsewhere [5]. Briefly, an Agilent 1200 module (Agilent, Palo Alto, CA,USA) equipped with a ZORBAX Eclipse XDB-C18 column(150 × 4.6 mm, 5 μm) was used for the chromatographic separation of the analytes with an injection volume of 20 μL. Mass spectrometric measurements were accomplished on an Applied Biosystems Sciex API 4000 system (Foster City, CA, USA) equipped with an electrospray ionization interface in negative ion mode. A binary gradient was used. The elution program was maintained at a flow rate of 1.0 mL min^{-1} starting with 63% water containing 1 mM ammonium acetate, which was increased to 95% methanol containing 1 mM ammonium acetate by 4.8 min and then decreased to 37%. The total elution time for the separation and elution of all the PFCAs was 8 min. Ions were acquired in multiple reaction monitoring (MRM) mode with a dwell time of ± 30 ms and with a unit mass resolution on both mass analyzers. An external calibration curve

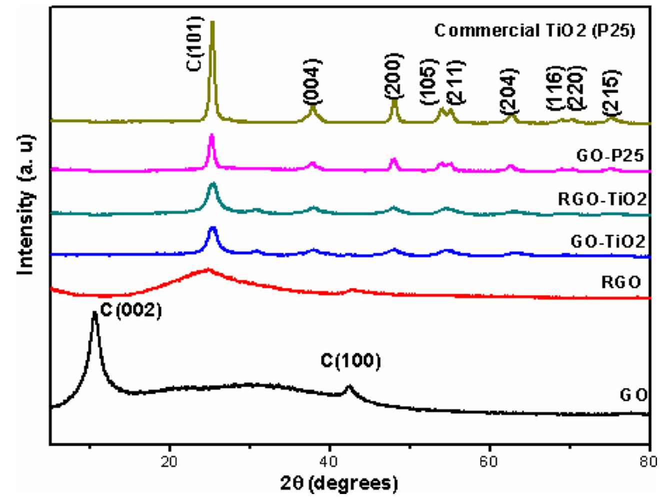

Fig. 1. XRD patterns of GO, RGO, TiO_2 (P25), and the TGNCs.

with a linear range from 0.1 to200 μgL^{-1} was used to quantify the analytes.

3. Results and discussion

3.1. Physical characterization

The XRD patterns of GO, RGO, and the TGNCs are shown in Fig. 1. In the case of GO, the typical (C002) peak appeared between the 2θ values 10° and 15°. Upon reduction of GO to form RGO, this peak shifted to higher angles with the appearance of a broad and low-intensity peak between the 2θ values 20° and 30°. The XRD peaks of the commercial TiO_2 are conveniently indexed to the anatase phase of TiO_2 using the JCPDS file of 21–1272 [33]. The diffraction patterns of all the TGNCs were nearly identical to one another and comparable to that of the commercial TiO_2 (P25), indicating that the synthesized TGNCs also exhibit the anatase phase of TiO_2. However, the diffraction peaks of layered GO disappeared from the patterns of the NCs, which might possibly be caused by overlap with the (101) peak of TiO_2 or by the small amounts of GO used.

The surface morphological characteristics of the P25, GO-P25, GO-TiO_2 and RGO-TiO_2 nanocomposites (TGNCs) and P25 were investigated by SEM, and the corresponding SEM images of the TGNCs are shown in Fig. 2 (a), (b), (c) and (d) respectively. Based on the SEM image of the TGNCs GO-P25, GO-TiO_2 nanocomposite shown in panel (b) and (c) of Fig. 2, the sample exhibits a rough surface morphology which differs from the pure P25. However, the layered structure related to GO is very difficult to see due to the small amount of GO used in the fabrication of the GO-TiO_2 nanocomposite. Similar observations were made for the RGO-TiO_2 nanocomposite sample, as shown in panel (d) of Fig. 2. The surface of this sample is also dominated by a rough morphology. The EDX profiles of P25, GO-P25, GO-TiO_2 and RGO-TiO_2 along with the weight and atomic percentages of the elements are shown in panels (a), (b), (c) and (d), respectively of Fig. 3. The elemental maps of Ti, O, and C shown in the EDX results confirms the presence of TiO_2 in the GO and RGO matrix.

To understand the morphology and crystallinity of the TGNCs, TEM analysis was performed, and the results are presented in Fig. 4. Panels of Fig. 4 show TEM images of the TiO_2, P25, GO, RGO, GO-TiO_2, RGO-TiO_2 and GO-P25 nanocomposites imaged at a scale of 200 nm. Spherical TiO_2 particles are seen to decorate the GO in the images. A similar morphology was observed for the other two TGNCs. Moreover, the HRTEM image shown in Fig. 4 (last panel) which indicates that the TiO_2 nanoparticles formed on GO have a perfect crystalline lattice with a d-spacing of 0.363 nm, corresponding to the (101) diffraction plane of

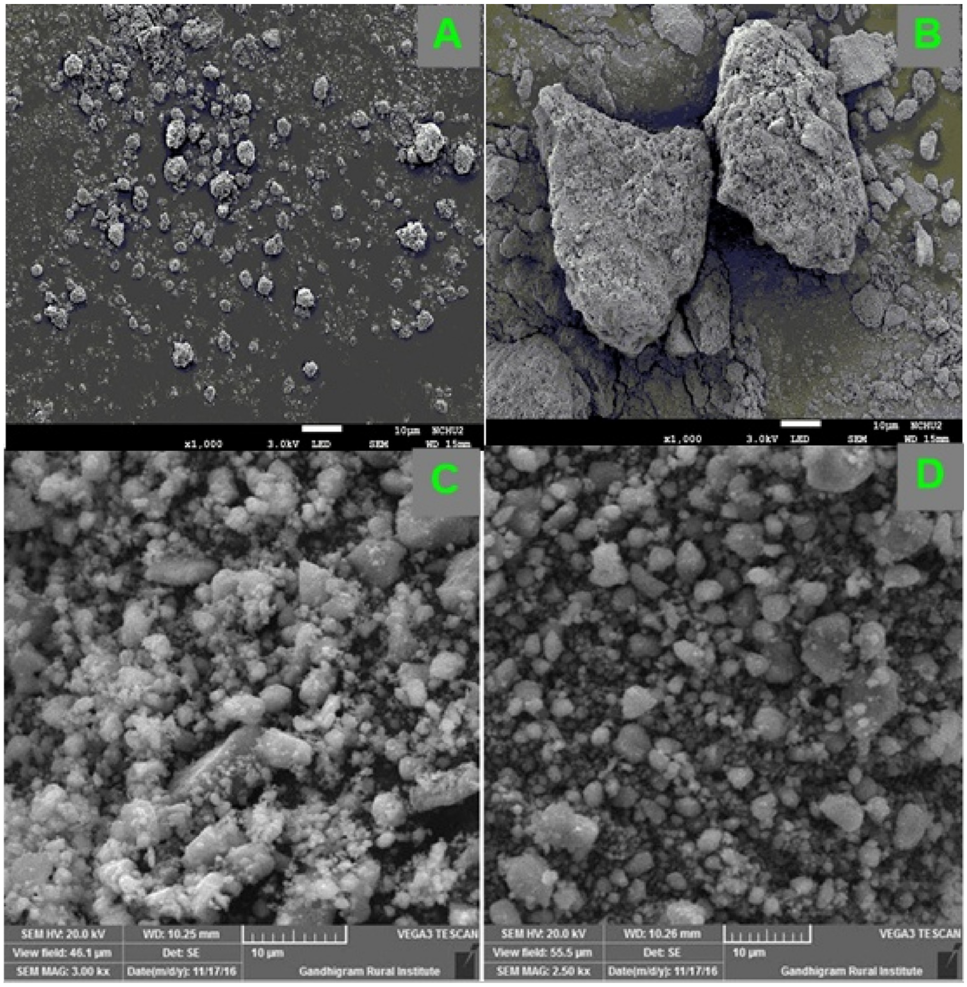

Fig. 2. SEM images of the (A) P25, (B) GO-P25, (C) GO-TiO_2 and (D) RGO-TiO_2 nanocomposites.

anatase TiO_2.

To assess the presence of GO and RGO in the nanocomposites synthesized in this study, X-ray photoelectron spectroscopy measurements are carried out and corresponding C 1 s and O 1 s XPS spectral patterns are presented in the panel (a) and (b), respectively of Fig. 5. The wide survey XPS and Ti 2p spectra of GO, GO-P25, GO-TiO_2 and RGO-TiO_2 are presented in the supplementary information Figure S3. The C 1 s of GO exhibits characteristic sp2 C—C and C—OH peaks at binding energies 285 and 286 eV, respectively consistent with the literature reports [34,35]. Interestingly, the intensity of C—C peak is significantly reduced when GO is mixed with commercial P25 and new additional peaks related to C—OH, C—O—C, C═O, HO—C═O are appeared at 286, 287, 288 and 289, respectively indicating the presence of GO in the GO-P25 nanocomposite. The decreased intensity of C—C peak is reasonably explained based on the fact that the TiO_2 content in the composite is higher when compared to GO. In this case, major amount of TiO_2 may cover the surface of GO and hence the C—C peak intensity is lowered. Similarly, the as-synthesized GO-TiO_2 too exhibits reduced C—C peak intensity when compared to the bare GO. However, in the case of GO-TiO_2 a new small peak appeared at lower B.E (282 eV) is attributed to the carbon bonded to titanium (C—Ti bond) [35,36]. When GO is partially reduced during the formation of RGO-TiO_2 the intensities of carbon bonds associated with oxygen is greatly reduced and only three types of carbon bonds that is C—C (slightly shifted to higher B.E, 286 eV), Ti-C (282 eV) and Ti-O-C (288.3 eV) indicating the presence of RGO with increased interactions of TiO_2 molecules [37].

The nature of chemical environment of oxygen in the GO, GO-P25, GO-TiO_2 and RGO-TiO_2 is assessed from O 1 s XPS patterns (Fig. 5b). The GO sample exhibit only one peak positioned at 533 eV corresponding to the oxygen single bonded to carbon [38]. In sample GO-P25, the O 1 s spectrum shows three bands. The peak appeared at lower B. E (at about 530 eV) can be assigned to the lattice oxygen of TiO_2 (O—Ti) consistent with the literature [39]. In addition, the peaks appeared at 532 and 533.4 eV can be assigned to the —OC and —OH groups, respectively [40]. A close inspection of O 1 s spectrum of GO-TiO_2 composite reveal that the peak corresponding to the O—Ti bond is shifted to higher B.E value compared to the GO-P25 sample suggesting the formation of a hydrogen bond between TiO_2 and GO [41]. In RGO-TiO_2 peaks related to O-Ti and O—C are retained where as the OH peak is completely removed indicating that GO is reduced to RGO.

The UV–vis diffuse reflectance absorption spectra for the three TGNCs along with TiO_2 and P25 was presented in the Fig. 6. The band gap of the samples were calculated from Kubelka-Munk plot. Accordingly, the band gaps of P25 and TiO_2 are 3.0 and 3.05 respectively. The addition of GO/RGO reduces the band gap which was calculated to be 2.6, 2.7 and 2.9 for GO-P25, GO-TiO_2 and RGO-TiO_2 respectively. The narrowing of the band gap as seen in the TGNCs allows for a more efficient utilization of the light than TiO_2 or P25 alone. The formation of Ti-C and Ti-O-C bonds are considered reason for the reduction of band gap energy in TGNCs [27,29].

3.2. Photocatalytic experimental results

3.2.1. Effect of the catalyst loading

The photocatalytic activity of various loadings of GO-TiO_2 (50 to 500 mg/L) in PFOA decomposition was studied, and the corresponding

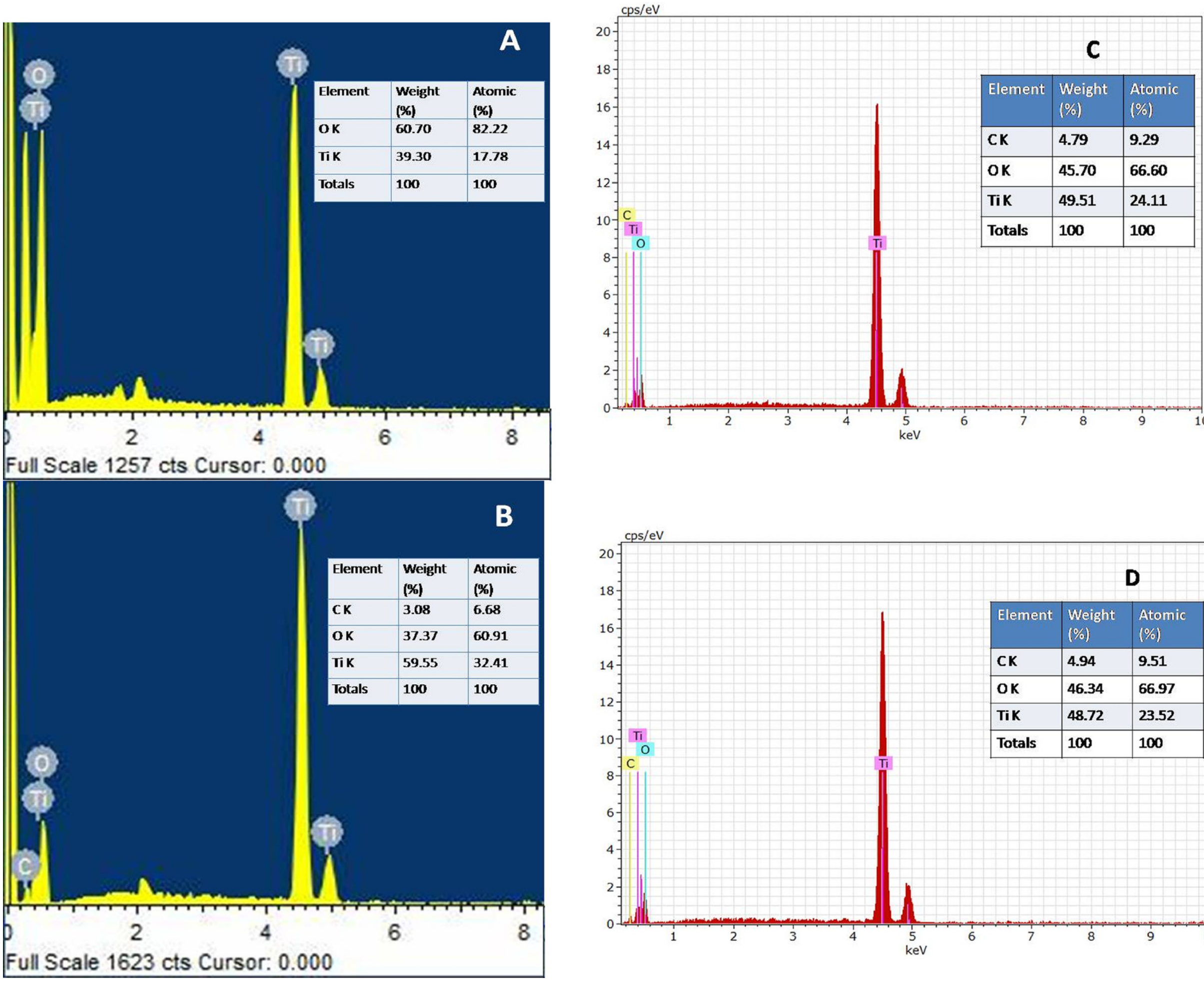

Fig. 3. EDX data for the (A) P25, (B) GO-P25, (C) GO-TiO_2 and (D) RGO-TiO_2 nanocomposites.

results are shown in Fig. 7 and Table 1. As the catalyst loading increased from 50 mg/L to 500 mg/L, the PFOA decomposition efficiency and the rate constants increased from 25% to 84% and from 0.031 to 0.238 h^{-1}, respectively, with the exception of the 200 mg/L GO-TiO_2 dose. The highest decomposition efficiency of 93.6% was achieved for the 200 mg/L GO-TiO_2 dose, which gave a rate constant of 0.288 h^{-1} and a half-life of 2.41 h. In general, with an increase in the catalyst loading, the number of sites available for photodecomposition increases, and hence, the decomposition efficiency is improved. However, a large amount of catalyst can result in agglomeration, thereby reducing the sites available for catalysis, as was observed in the case of catalyst loadings above 200 mg/L. Therefore, a catalyst dose of 200 mg/L was found to be optimum under the studied experimental conditions and was employed in further photocatalytic experiments. In order to check the reusability of the catalyst material, an experiment was conducted with GO-TiO_2 as the photocatalyst. All other experimental conditions being same, in the case of the reuse of the catalyst conducted only once, the photocatalyst was separated from the reaction solution through centrifugation and then washed with both methanol and water several times before drying in oven for 3 h at 200 °C. Upon reuse of the catalyst the PFOA decomposition efficiency was found to be 78% which means the photocatalytic efficiency reduced by approximately 15%.

3.2.2. Comparison of the catalysts

Comparative experiments were conducted to investigate the photocatalytic activities of the various TGNCs in PFOA decomposition. The TGNCs studied in addition to Degussa P25 (P25 alone) and GO-TiO_2 are GO-P25 and RGO-TiO2, and the corresponding results are shown in Fig. 8. Direct photolysis of PFOA by 254 nm UV radiation is negligibly slow in our experiments, as was previously reported by several studies [19,22,42]. Under similar experimental conditions of 12 μM PFOA and 200 mg/L catalyst, only 16% of the PFOA was decomposed after 8 h by P25 alone, whereas in the presence of GO, the TiO_2 catalyst, the PFOA decomposition efficiency was markedly enhances by nearly 6 times. The PFOA decomposition efficiencies of all three TGNCs are nearly the same, with values above 90%. Among the TGNCs, RGO-TiO_2 exhibited a higher performance (99.2%) than the other two (GO-TiO_2: 93.6% and GO-P25: 96%), which had comparable performances. Furthermore, the result for RGO-TiO_2 is consistent with that of Beatriz, et al. (2017) [30], who reported a 93 ± 7% decomposition efficiency. The BET specific surface area of RGO-TiO_2 was found to be 115.5 m^2/g which is more than twice as compared to that of P25 which is 50 m^2/g. The BET surface area as observed in the present study is higher when compared to the previous study that employed reduced graphene oxide titanium dioxide (62.2 m^2/g) catalyst for the decomposition of PFOA [30]. The pore volume of RGO-TiO_2 TGNC was found to be 0.239 cm^3/g and mean pore diameter was 8.273 nm. The BET surface area for GO-TiO_2 was 121.3 m^2/g. These similar specific surface areas might have eventually resulted in the similarities of TGNCs decomposition efficiencies. The morphology and structure of the TGNCs also influence the photocatalytic decomposition efficiency. Notably the spherical TiO_2 nanoparticles tend to agglomerate [23,43,44], whereas a homogeneous dispersion of TiO_2 particles on the GO/RGO sheet (Figs. 3 and 4) bereft TGNCs from TiO_2 particle agglomeration benefitting photocatalytic treatment process with an increased utilization of UV light.

The enhancement of the decomposition efficiency in the presence of

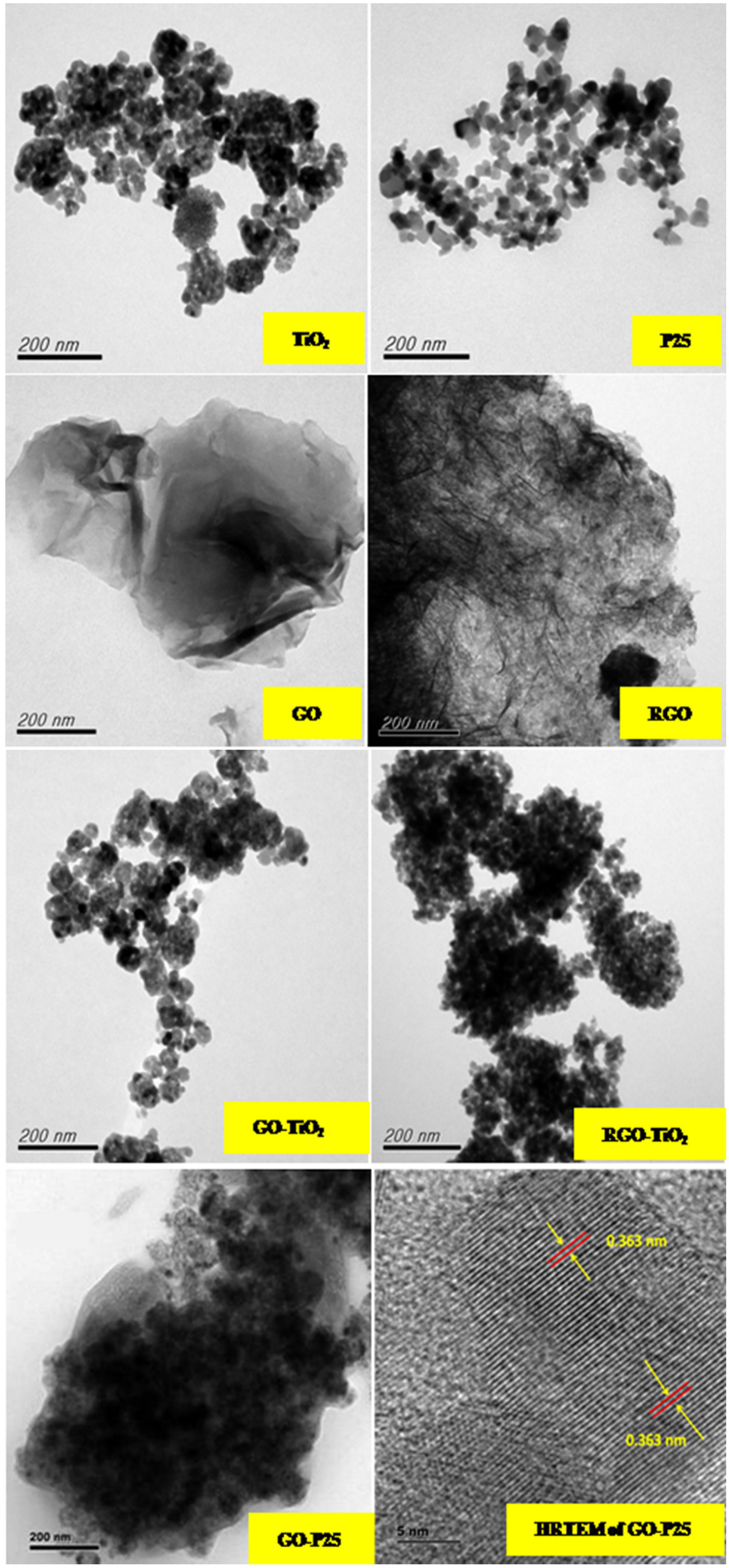

Fig. 4. TEM images of TiO_2, P25, GO, RGO, GO-TiO_2, RGO-TiO_2 and GO-P25 nanocomposite at a 200 nm length scale including HRTEM image of GO-P25.

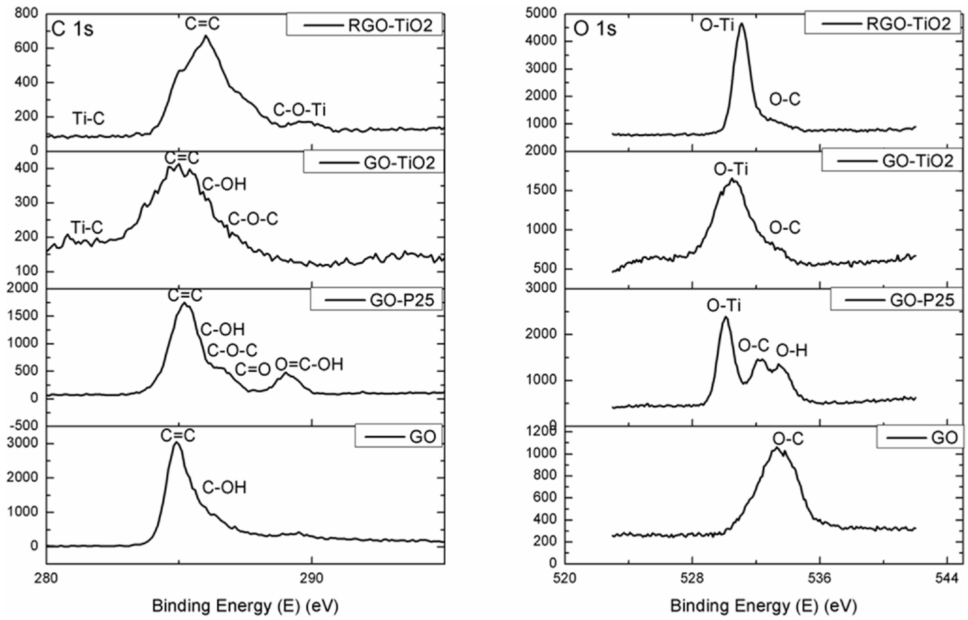

Fig. 5. X-ray photoelectron spectral patterns (a) C 1 s and (b) O 1 s of GO, GO-P25, GO-TiO_2 and RGO-TiO_2.

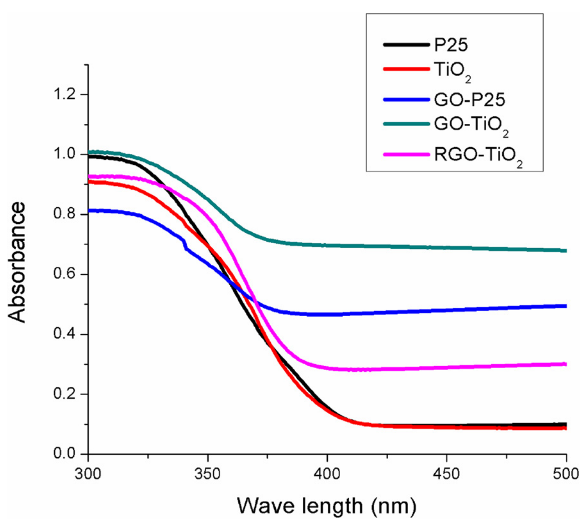

Fig. 6. Diffuse reflectance absorption spectra of P25, pure TiO_2, and three TGNCs (GO-P25, GO-TiO_2, and RGO-TiO_2).

GO or RGO is seemingly due to the charge-carrying capacity of graphene. In particular, the as-prepared TGNCs used in this study possibly possess chemical bonds unlike those in a simple mixture of GO or RGO and TiO_2, resulting in intimate interactions between the TiO_2 particles and graphene sheets. These intimate Ti-C chemical bonds formed in the TGNCs (as evidenced from the XPS data, Fig. 5) during the hydrothermal synthesis process are expected to effectively facilitate interfacial electron transfer, which aids in electron-hole separation [28]. The enhanced electron-hole separation seems to facilitate the participation of a larger number of holes in the photooxidation process in the present study and hence enhances the PFOA photodecomposition efficiency. An addition of GO or RGO into TiO_2 based nanocomposites offers to reduce the band gap energy when compared to TiO_2 alone. The reduction in the band gap energy and good transparency offered by one-atom thick GO or RGO enable the TGNCs for the effective utilization of irradiated light [28,30,45].

The PFOA decomposition kinetics were studied by fitting the data to the pseudo-first-order kinetic model (Table 1).The corresponding rate constants of P25, RGO-TiO_2, GO-TiO_2 and GO-P25 are 28.8, 0.45, 0.288 and 0.270 h^{-1}, respectively. The higher performance of RGO-TiO_2 relative to that of the other two catalysts with GO probably results from the decrease in the number of oxygen-containing functional groups on the GO surface as a result of the reduction process, as these groups might compete for the radicals generated in the photocatalytic system. Therefore, the lower number of oxygen-containing functional groups on the RGO surface (as can be observed from Fig. 5(a)) resulted in a decrease in the half-life value of PFOA decomposition from 2.5 to 1.5 h.

3.2.3. Effects of quenching agents

The photocatalytic decomposition of PFOA by the TGNCs was studied in the presence of different quenching agents under oxygen atmosphere. PFOA decomposition in a heterogeneous photocatalytic system, particularly that employing TiO_2 as a photocatalyst, may involve three processes [12]. First, an electron in the conduction band of excited TiO_2 moves to the surface of the catalyst, where it reacts with adsorbed water to form hydrated electrons. These hydrated electrons can attack the adsorbed PFOA molecule to form shorter-chain compounds [11,12]. Second, PFOA decomposition can occur through the hole generated in the valence band. The hole directly accepts an electron from a PFOA anion, resulting in the formation of a perfluorocarboxy radical [45]. Third, the valence band hole can consume an electron from a water molecule to generate a hydroxyl radical, which subsequently consumes an electron from a PFOA anion; this is an indirect pathway through the valence band junction [33]. Although all of these processes can occur simultaneously, the predominant path decides the decomposition efficiency and overall kinetics of the photocatalytic system. In the present study, two quenching agents, an oxidant (sodium persulfate) and a reductant (isopropyl alcohol), were introduced into the reaction mixture to determine the predominant PFOA decomposition mechanism (Fig. 9). Interestingly, the PFOA decomposition efficiency increased in the presence of both the oxidizing and reducing agents. However, a higher decomposition efficiency was achieved in the presence of isopropyl alcohol (Fig. 9).

Sodium persulfate is a strong oxidant that quenches conduction band electrons. Upon the addition of 2.1 mM sodium persulfate (400 mg), the PFOA decomposition efficiency increased from 57.2% to 76.7%. The sulfuric acid radical anion produced by the photolysis of persulfate is known to react with PFOA to accomplish PFOA degradation [42–44,46]. In a photochemical system, persulfate exhibits strong absorption in the UV region, which generates sulfuric acid radicals that

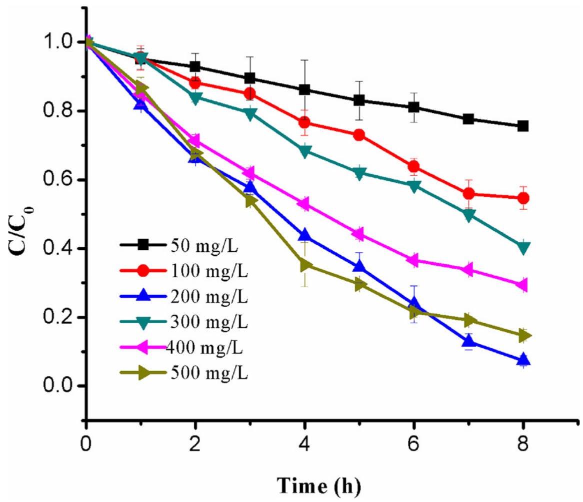

Fig. 7. Effect of the GO-TiO_2 catalyst loading on PFOA decomposition ([PFOA] = 12 µM, volume of reaction solution = 50 mL, O_2 flowrate = 5cc/min, Initial pH = 4.64–4.95).

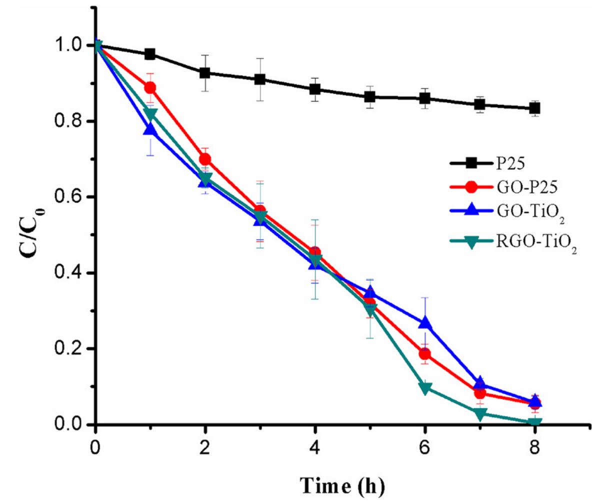

Fig. 8. Effects of P25 and the three different TGNCs (GO-P25, GO-TiO_2 and RGO-TiO_2) on PFOA decomposition ([PFOA] = 12 µM, catalyst load = 200 mg/L, volume of reaction solution = 50 mL, O_2 flowrate = 5cc/min, Initial pH = 4.64–4.95).

can perform PFOA decomposition. However, in a heterogeneous photocatalytic system, TiO_2 absorbs more energy than persulfate, and hence, the role of persulfate is more that of an electron acceptor [20,46]. In addition, the concentration of persulfate used in this study (2.1 mM) was lower than that used in other PFOA decomposition studies (10–50 mM) [44]. The aqueous electrons generated during photocatalysis can also directly decompose PFOA, but this mechanism appears to be more dominant in nitrogen atmospheres, which favor reductive decomposition [11]. In the present experiments, the reaction mixture was purged with oxygen gas, which minimized the involvement of aqueous electrons in the decomposition of PFOA. Therefore, persulfate can be concluded to increase the PFOA decomposition efficiency over that of the system with GO-TiO_2 alone by acting as an electron acceptor and preventing electron-hole pair recombination in addition to performing the direct oxidation of PFOA.

Isopropyl alcohol is a strong reducing agent that annihilates hydroxyl radicals and other oxidizing substances in the bulk solution [26]. Interestingly, when 1.6 mM isopropyl alcohol (96 mg) was introduced into the reaction mixture, the PFOA decomposition efficiency drastically increased relative to that in the absence of a quenching agent from 57.2% to 96.1%.

During the photocatalytic decomposition of PFOA by TiO_2 or metal-modified TiO_2, hydroxyl radicals usually either accept electrons from ionized PFOA [23,24] or assist in the conversion of the unstable alcohol formed during the photo-Kolbe reaction to shorter-chain compounds [19,22,25,30]. However, in the presence of isopropyl alcohol, these hydroxyl radicals in the bulk solution are scavenged, facilitating the direct conversion of PFOA anions to PFOA radicals at the photoholes on TiO_2. This situation enhances the decomposition of PFOA.

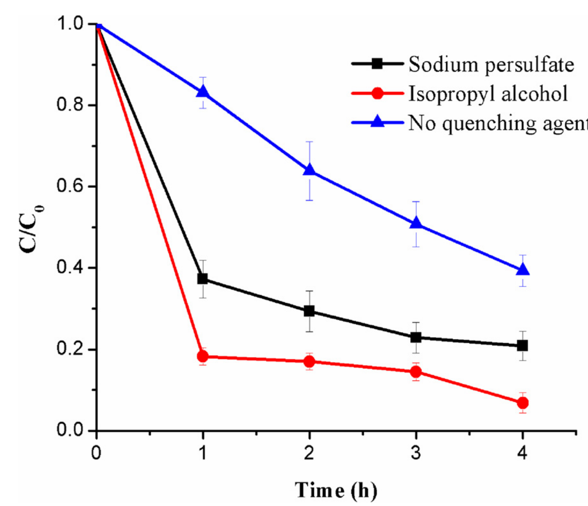

Fig. 9. Effects of quenching agents on the photocatalytic decomposition of PFOA by GO-TiO_2 ([PFOA] = 12 µM, catalyst load = 200 mg/L, Isopropyl alcohol = 1.6 mM, Sodium persulfate = 2.1 mM, volume of reaction solution = 50 mL, O_2 flowrate = 5cc/min, Initial pH = 4.64–4.95).

Since isopropyl alcohol is a reducing agent, we could speculate that PFOA decomposition occurred through the reductive pathway, as was observed by [11] upon the addition of hole scavengers such as oxalic acid and potassium iodide. However, the experimental conditions of that study, such as the lower pH (2.47) and nitrogen atmosphere, which

Table 1
Photocatalytic decomposition of PFOA by the TGNCs and the corresponding kinetic rate constants.

S.No	Initial PFOA µM	Catalyst Material	Catalyst Load mg/L	PFOA Decomposition Efficiency %	Half Life H	Pseudo-First-Order Rate Constant h^{-1}	R^2Value
1	12	GO-TiO_2	50	25	22.3	0.031	0.928
2	12	GO-TiO_2	100	43	10.2	0.068	0.975
3	12	GO-TiO_2	200	93.6	2.41	0.288	0.924
4	12	GO-TiO_2	300	59	7.00	0.098	0.968
5	12	GO-TiO_2	400	70	4.40	0.159	0.991
6	12	GO-TiO_2	500	84	2.90	0.238	0.986
7	12	GO-P25	200	96	2.56	0.270	0.919
8	12	RGO-TiO_2	200	99.2	1.54	0.45	0.814
9	12	P25	200	16	28.8	0.024	0.837

usually favor reductive decomposition, differed from those of the current study. In addition, in a photocatalytic experiment with TiO_2 as the photocatalyst and an external oxygen supply, Sansetora et al. [30] reported that PFOA was decomposed into shorter-chain compounds through the transfer of an electron from a PFOA anion to a hole on TiO_2, thereby generating a PFOA radical. In their study, the conduction band electrons were reported to engage in the conversion of oxygen to superoxide radicals, which participated in the propagation of the decomposition reaction. Superoxide radicals can reductively decompose PFOA, as reported by da Silva-Rackov et al. [32]. However, in the case of the GO-TiO_2 catalyst, the electrons partially sink into the GO sheet, preventing electron-hole recombination. In such a case, the formation of superoxide radicals on TiO_2 is relatively slow in comparison to the conversion of an adsorbed PFOA anion to a radical, thereby more strongly facilitating the oxidative pathway than the reductive pathway. Thus, based on the results obtained in this study, when isopropyl alcohol was added to the reaction, it scavenged the bulk hydroxyl radicals, thereby facilitating the direct conversion of PFOA anions to PFOA radicals and leading to a higher decomposition efficiency. Moreover, a faster (90% in 1 h) and higher decomposition efficiency (100% in 4 h) was observed when 0.32 M isopropyl alcohol was added, affirming the discussed results.

3.2.4. Decomposition intermediates

The shorter-chain intermediates generated during the photocatalytic decomposition of PFOA were quantified by LC–MS/MS. The time-dependent generation of shorter-chain compounds is shown in Fig. 10a, b and c for GO-P25, GO-TiO_2 and RGO-TiO_2, respectively. In all cases, the concentrations of the shorter-chain PFCAs increased with increasing reaction time except for that of PFHpA, which showed an initial increase followed by a gradual decrease. The appearance of shorter-chain PFCAs was always preceded by the observation of their longer-chain analogs in all cases. Thus, the order of the observed PFCA concentrations was as follows: PFHpA > PFHxA > PFPeA > PFBA > PFPrA. TFA was not detected, probably because its concentration was too low. This result indicates that stepwise decomposition of PFOA occurred to form shorter-chain PFCAs via the removal of each CF_2 unit. Additionally, consistent with the rate constants and catalytic performance, the RGO-TiO_2 catalyst generated larger quantities of the intermediates more quickly than GO-TiO_2, which was followed by GO-P25. Therefore, the overall performance of the TGNCs exhibited the trend RGO-TiO_2 > GO-TiO_2 > GO-P25.

3.2.5. Mechanism of decomposition

The reported energy levels of the valence band (VB) and conduction band (CB) of TiO_2 materials are 2.81 and -0.39 V vs normal hydrogen electrode (NHE), respectively, whereas the calculated Fermi energy level of graphene is − 0.08 V vs NHE; therefore, the transfer of a photogenerated electron inthe CB of TiO_2 to graphene is feasible [29]. Consequently, GO acts as a sink for the photogenerated electrons.

$$TiO_2 + h\nu \rightarrow e- + h^+(VB) \tag{1}$$

$$GO + e-(CB) \rightarrow e-(GO) + TiO_2 \tag{2}$$

Based on the influence of the quenching agents in this study and the literature on the photodecomposition of PFOA (Fig. 11), PFOA decomposition may occur simultaneously by direct electron transfer from the dissociated PFOA to the VB of TiO_2, generating a perfluoroperoxy radical (Eq. (3)), which is highly unstable and thus spontaneously undergoes photo-Kolbe-like decarboxylation to form a peralkyl radical (Eq. 4). In another pathway, the attack of a PFOA anion by a hydroxyl radical (Eq. 5) forms a peralkyl radial via cleavage of the C—C bond between the perfluorinated alkyl chain and the carboxylic group.

$$F(CF_2)_nCOO- + h^+ \rightarrow F(CF_2)_nCOO^{\bullet} \tag{3}$$

$$F(CF_2)_nCOO^{\bullet} \rightarrow F(CF_2)_n^{\bullet} + CO_2 \tag{4}$$

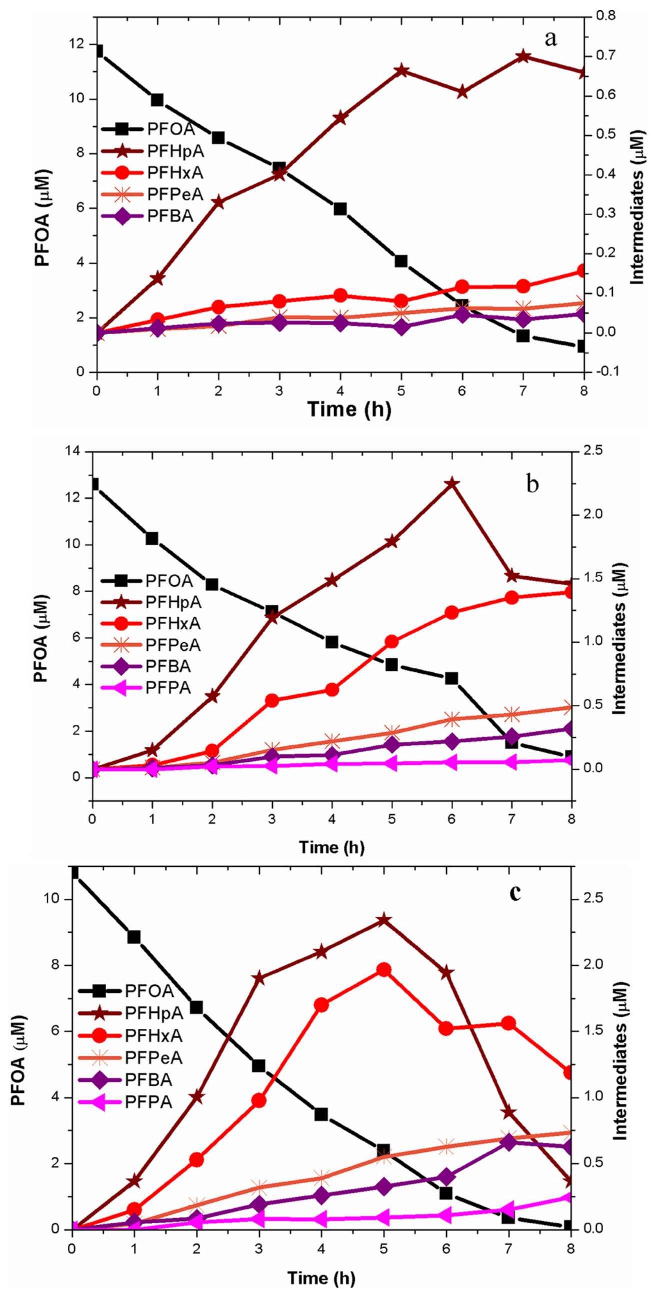

Fig. 10. Shorter-chain intermediates generated during the photocatalytic decomposition of PFOA by TGNCs (a) GO-P25, (b) GO-TiO_2 and (c) RGO-TiO_2 ([PFOA] = 120 μM, catalyst load = 200 mg/L, volume of reaction solution = 50 mL, O_2 flowrate = 5cc/min, Initial pH = 4.64–4.95).

$$F(CF_2)_nCOO- + {}^{\bullet}OH \rightarrow {}^{\bullet}F(CF_2)_nCOO^+ + OH- \tag{5}$$

$${}^{\bullet}F(CF_2)_nCOO^+ \rightarrow F(CF_2)_n^{\bullet} + CO_2 + H^+ \tag{6}$$

The peralkyl radicals formed by Eq. (4) or 6 react with water to form $F(CF_2)_nOH$ (Eq. (7)). This thermally unstable alcohol forms acid fluoride (Eq. (8)), which is later hydrolyzed to form perfluorocarboxylic acid with one fewer CF_2 unit (Eq. (9)).

$$F(CF_2)_n^{\bullet} + H_2O \rightarrow F(CF_2)_nOH + H^+ \tag{7}$$

$$F(CF_2)_nOH \rightarrow F(CF_2)_{n-1}COF + F^- + H^+ \tag{8}$$

$$F(CF_2)_{n-1}COF + H_2O \rightarrow F(CF_2)_{n-1}COOH + F^- + H^+ \tag{9}$$

These reactions continue until the mineralization of PFOA and the generated shorter-chain intermediates is complete. Based on the effects

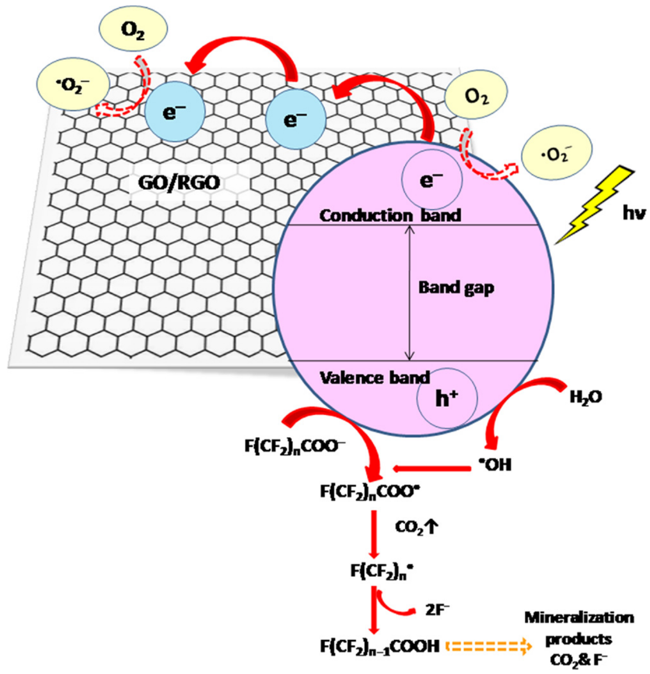

Fig. 11. Mechanism of PFOA decomposition by photocatalytic treatment with the TGNCs.

of the quenching agents on PFOA decomposition by the TGNCs, the direct conversion of PFOA at the VB of TiO_2 is rate limiting, and once the barrier is crossed, the decomposition efficiency increases enormously (upon the addition of isopropyl alcohol). Wang and Zhang [11] mentioned that the photocatalytic degradation efficiency depends on the quantity and activity of photogenerated electron-hole pairs. Since the photogenerated electron-hole pair plays the largest role in the photocatalytic system and since PFOA decomposition is limited by direct PFOA conversion at the VB hole of TiO_2, the present study effectively prevented electron-hole pair recombination by the use of the TGNCs.

4. Conclusions

The TiO_2-graphene nanocomposites GO-P25, GO-TiO_2 and RGO-TiO_2 were synthesized by a facile hydrothermal process and then successfully characterized. The TGNCs exhibited high photocatalytic performance, decomposing more than 90% of the intractable PFOA, while only 16% of the PFOA was decomposed by P25 alone. The improved performance of the TGNCs relative to that of TiO_2 alone is due to the charge separation ability of graphene, which acted as an electron sink. Among the studied TGNCs, RGO-TiO_2 (200 mg/L) exhibited superior performance in terms of both the decomposition efficiency (99.2%) and the kinetics (0.45 h^{-1}). The observed higher performance of the RGO-TiO_2 relative to that of the other two catalysts with GO likely results from the decrease in the number of oxygen-containing functional groups on the GO surface caused by the reduction process, as these groups might compete for the radicals generated in the photocatalytic system. Two quenching agents, sodium persulfate and isopropyl alcohol, were added to the photocatalytic system, and in both cases, an increase in the PFOA decomposition efficiency was observed. Persulfate acted as an electron acceptor in the present photocatalytic system, thus preventing electron-hole pair recombination in addition to directly oxidizing PFOA. Isopropyl alcohol scavenged the bulk hydroxyl radicals, thereby facilitating the direct conversion of PFOA anions to PFOA radicals and leading to a higher decomposition efficiency. According to the effects of the quenching agents on PFOA decomposition by the TGNCs, direct conversion of PFOA at the VB of TiO_2 is rate limiting, and once the barrier is crossed, the decomposition efficiency increases enormously. The TGNCs could efficiently decompose PFOA into shorter-chain perfluorocarboxylic acids. The present work demonstrates the facile synthesis of TGNCs with high photocatalytic performance, as illustrated in the decomposition of PFOA. An understanding of the mechanism of photocatalytic PFOA decomposition using the TGNCs as photocatalysts was also obtained.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jece.2018.10.003.

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